AIR UNIVERSITY
Department of Cyber Security — Digital Forensics Laboratory | Course CY334

EVIDENCE CHAIN OF CUSTODY RECORD

Digital Forensic Examination — Labs 1 through 5

FORM NO.	CASE NUMBER	PAGE STATUS
COC-CY334-001	**CASE-2026-M3**	**ACTIVE — LAB 5 COMPLETE**

SECTION A — CASE & INVESTIGATOR INFORMATION

CASE NUMBER CASE-2026-M3	CASE NAME Hard Disks & File Systems — Disk Image Forensic Analysis
INVESTIGATION TYPE Digital Forensics — Academic Laboratory Exercise	REPORTING INSTITUTION Air University, CY334 Digital Forensics
LEAD INVESTIGATOR Abdul Wahid	INVESTIGATOR ID / ROLL NO. 243699
DATE CASE OPENED 30 September 2026	DATE OF THIS RECORD 6 October 2026

This record documents the receipt, hashing, examination, recovery, and export of digital evidence associated with the above case, in accordance with ISO/IEC 27037, NIST SP 800-86 and standard evidence-handling procedure (Labs 1–5).

SECTION B — EVIDENCE DESCRIPTION LOG

Item No.	Evidence ID	Description	Source / Derived From	Date/Time Created	Condition
1	EV-001	Linux_Evidence_001.img — Ext4 volume, Ubuntu 16.04 VM "roger-Virtual-Machine", 4,145,336,320 bytes (incomplete)	Supplied image (Linux workstation, crime scene)	29 Sep 2026, 10:41 †	Verified (hashed)
2	EV-002	Windows_Evidence_001.dd — bare NTFS volume "Evidence", 2,147,483,648 bytes	Supplied image (organization's security team)	1 Oct 2026, 10:31 †	Verified (hashed)
3	EV-003	Windows_Evidence_002.dd — MBR disk, two NTFS partitions, 26,843,545,600 bytes	Supplied image (organization's security team)	1 Oct 2026, 10:32 †	Verified (hashed)
4	EV-002-RECOVERED	Recovered_EV002 — 22 deleted files recovered by metadata (tsk_recover)	EV-002	3 Oct 2026, 15:52	Verified (hashed)
5	EV-001-CARVED-WH	Carved_WinHex_EV001 — WinHex carving output (232 files + log); carved_inventory.csv	EV-001	4 Oct 2026, 02:00	Inventory hashed
6	EV-001-CARVED-PR	PhotoRec_EV001_Free and _Whole — PhotoRec carving (31 files saved per run)	EV-001	4 Oct 2026, 11:26	Verified (hashed)
7	EV-001-SLACK	slack_HideMarkedCols.bin — 3,584 bytes of file slack (inode 46066), all 0x00	EV-001	4 Oct 2026, 12:19	Verified (hashed)
8	EV-TIMELINES	Lab4 — body files and mactime timelines, EV-001 and EV-002	EV-001, EV-002, EV-003	4 Oct 2026, 03:42	Verified (hashed)
9	EV-SIGCHECK	Lab5\sigcheck_EV002.csv — signature test of the 22 files; times_check.csv	EV-002-RECOVERED	4 Oct 2026, 20:12	Verified
10	EV-INVALID	INVALID_wrong_source_* — PhotoRec output from the examiner's own drives, not evidence	Wrong source (PhysicalDrive0/1)	4 Oct 2026, 03:20	Quarantined

Full technical description, hash values, and forensic tool versions are maintained in the associated laboratory report (Module 3, Labs 1–5, v2.0) and are incorporated herein by reference. † Creation time of the copy on the examiner drive: Explorer showed 29 Sep 2026, 10:41:30 for EV-001, but a 6 Oct 2026 PowerShell listing showed 1 Oct 2026 (10:30–10:32) for all three images. Modified times (27 Oct 2023) and all hashes are unchanged; the reason for the later creation time was not recorded.

SECTION C — CHAIN OF CUSTODY — TRANSFER RECORD

Every change of possession, location, or custodial responsibility for an evidence item must be recorded below with both a releasing and a receiving signature. An unexplained gap in this log constitutes a break in chain of custody.

Item	Date/Time	Released By / From	Received By / To	Purpose of Transfer
EV-001	29 Sep, 10:41	Lab-supplied evidence media	Examiner drive Crack-X (E:), CRACK-S25	Receipt; image copied to Tools and Forensic Images †
EV-001	30 Sep, 20:05	Examiner drive (read-only)	Autopsy case CASE-2026-M3 (C:\Users\Administrator\Desktop)	Case created; EV-001 loaded for Lab 1 analysis
EV-001	1 Oct, 10:30	Lab-supplied evidence media	Examiner drive Crack-X (E:), CRACK-S25	Copy creation time in 6 Oct listing †
EV-002	1 Oct, 10:31	Lab-supplied evidence media	Examiner drive Crack-X (E:), CRACK-S25	Receipt; image copied to Tools and Forensic Images †
EV-003	1 Oct, 10:32	Lab-supplied evidence media	Examiner drive Crack-X (E:), CRACK-S25	Receipt; image copied to Tools and Forensic Images †
EV-001 to EV-003	[baseline date]	Examiner drive (images)	Examiner drive (images)	Baseline hashing (certutil + Get-FileHash)
EV-002-RECOVERED	3 Oct, 15:52	EV-002 (read-only)	Exports\Recovered_EV002	Deleted-file recovery, tsk_recover (Lab 2)
EV-001-CARVED-WH	4 Oct, 02:00	EV-001 (read-only)	Exports\Carved_WinHex_EV001	WinHex carving, read-only mode, 02:01–02:13 (Lab 3)
EV-INVALID	4 Oct, 03:20	Examiner's physical drives	Exports\INVALID_wrong_source_*	PhotoRec run on wrong source; folders renamed, kept, excluded (03:20–04:04)
EV-TIMELINES	4 Oct, 03:42	EV-001, EV-002 (read-only)	Exports\Lab4	Body files and mactime timelines (Lab 4), 03:53–04:30
EV-001-CARVED-PR	4 Oct, 11:26	EV-001 (read-only)	Exports\PhotoRec_EV001_Free and _Whole	PhotoRec carving, free-space and whole-disk runs (Lab 3)
EV-TIMELINES	4 Oct, 12:08	EV-001 (read-only)	Exports\Lab4	EV-001 timeline CSV rebuilt; earlier empty timeline discarded
EV-001-SLACK	4 Oct, 12:19	EV-001 (read-only)	Exports\Lab3	File-slack extraction, 3,584 bytes
EV-SIGCHECK	4 Oct, 20:12	EV-002-RECOVERED	Exports\Lab5	Signature test of 22 recovered files (Lab 5)
EV-SIGCHECK	6 Oct, 14:01	EV-002-RECOVERED	Exports\times_check.csv	Timestamp-order check
EV-001 to EV-003	[closing date]	Examiner drive (images)	Examiner drive (images)	Closing SHA-256 re-verification (all match baseline)

Rel. = Released by (transferring custodian). Rec. = Received by (accepting custodian). Where the same individual performs both roles, this reflects continuous sole-custodian handling within a controlled academic lab environment and is documented as such. Dates and times are workstation local time (UTC+5). Items in brackets are to be completed by the examiner.

SECTION D — EVIDENCE STORAGE & ACCESS CONTROL LOG

Item	Storage Location	Access Control	Integrity Control	Modified?
EV-001	E:\DF Lab 3\Lab 3\Tools and Forensic Images\	Examiner workstation CRACK-S25; read-only (WinHex write-protected, PhotoRec RO)	MD5 + SHA-256 baseline; re-hashed at close; last-write time 27 Oct 2023	No
EV-002	E:\DF Lab 3\Lab 3\Tools and Forensic Images\	Examiner workstation; read-only	MD5 + SHA-256 (certutil and Get-FileHash); last-write time 27 Oct 2023	No
EV-003	E:\DF Lab 3\Lab 3\Tools and Forensic Images\	Examiner workstation; read-only	MD5 + SHA-256 (certutil and Get-FileHash); last-write time 27 Oct 2023	No
EV-002-RECOVERED	E:\DF Lab 3\Lab 3\Exports\Recovered_EV002\	Separate output folder; image never written	MD5 list of 22 files exported	No
EV-001-CARVED-WH	E:\DF Lab 3\Lab 3\Exports\Carved_WinHex_EV001\	Separate output folder; image never written	carved_inventory.csv (size, first 8 bytes, MD5 of 233 records)	No
EV-001-CARVED-PR	E:\DF Lab 3\Lab 3\Exports\PhotoRec_EV001_Free\ and \PhotoRec_EV001_Whole\	Separate output folder; image never written	Image re-hashed before and after runs; per-file MD5	No
EV-001-SLACK	E:\DF Lab 3\Lab 3\Exports\Lab3\	Separate output folder	SHA-256 + MD5 of extracted file	No
EV-TIMELINES	E:\DF Lab 3\Lab 3\Exports\Lab4\	Separate output folder	MD5 of body files and timelines (certutil + Get-FileHash)	No
EV-SIGCHECK	E:\DF Lab 3\Lab 3\Exports\Lab5\	Separate output folder; recovered files read as bytes only, never opened	Source files hashed (MD5) before testing	No
EV-INVALID	E:\DF Lab 3\Lab 3\Exports\INVALID_wrong_source_*\	Quarantined; excluded from all findings	Renamed with INVALID_ prefix; not deleted	Renamed only
CASE-2026-M3 (Autopsy)	C:\Users\Administrator\Desktop\CASE-2026-M3	Examiner account; case database	Created 30 Sep 2026, 20:05; last write 4 Oct 2026, 01:53	Yes — working file

SECTION E — EVIDENCE INTEGRITY VERIFICATION (HASH LOG)

Item	Algorithm	Value	Verified By	Date
EV-001	MD5	*00ccccb1628d59e2170872e9cfd57b4d*	Abdul Wahid	2 Oct 2026
EV-001	SHA-256	*e198ac948d5a3bae08f6a1a83f1f19ff219312bf53a5e809fc1db23214a6f87d*	Abdul Wahid	2 Oct 2026
EV-002	MD5	*136d87d040222292f3a91746b49a90d2*	Abdul Wahid	2 Oct 2026
EV-002	SHA-256	*eeef3666ab1fda6b7d0af6bf2939eabe6864d824a36d6b109ee70f9cb23ba5a7*	Abdul Wahid	2 Oct 2026
EV-003	MD5	*eb1fe400707af8dc50d3ed866841630c*	Abdul Wahid	2 Oct 2026
EV-003	SHA-256	*28d7ba346c828ec7ea8bacd17473a69cd6781bb1292a951f734b6a050bf600b3*	Abdul Wahid	2 Oct 2026
EV-001 to EV-003	SHA-256	Closing re-hash of all three images equals the baseline values above	Abdul Wahid	6 Oct 2026
EV-001 (HideMarkedCols.xls)	MD5	*51ae5c0f8442964147f11c04676a411e*	Abdul Wahid	4 Oct 2026
EV-001 (crontab)	MD5	*8f111d100ea459f68d333d63a8ef2205*	Abdul Wahid	[Lab 1 date]
EV-002-RECOVERED (Word_Doc1.docx)	MD5	*eabeb38580705f67e46da01b8d311897*	Abdul Wahid	3 Oct 2026
EV-002-RECOVERED (Word_Doc.docx)	MD5	*0fb522ab91b960ad9f9c27095e01e2e2*	Abdul Wahid	4 Oct 2026
EV-001-SLACK	MD5	*b4202f7fe985b9648b4676e6f70832bd*	Abdul Wahid	4 Oct 2026
EV-TIMELINES (EV-002 body file)	MD5	*f369c33f1d2b1739b29f44b1b50d1cce*	Abdul Wahid	4 Oct 2026
EV-TIMELINES (EV-002 timeline)	MD5	*715b95a2ca8dbe4030646fa57dd2a70f*	Abdul Wahid	4 Oct 2026
EV-TIMELINES (EV-001 body file)	MD5	*2f53c1e80158872db2616ccd829f95bc*	Abdul Wahid	4 Oct 2026
EV-TIMELINES (EV-001 timeline)	MD5	*59538b7870043977e5e71e9b340d8765*	Abdul Wahid	4 Oct 2026
EV-002-RECOVERED	Hash list	MD5 list of the 22 recovered files (CSV, Exports\)	Abdul Wahid	3 Oct 2026

EV-001 was hashed twice with certutil; EV-002 and EV-003 were each hashed with Get-FileHash and certutil. All results agreed. Dates for derived items are taken from file creation times in the Exports folder; items in brackets are to be completed by the examiner.

SECTION F — FINAL DISPOSITION OF EVIDENCE

☒ **Retained** in evidence storage pending further examination (later modules — registry, event-log and timestamp-consistency analysis)

☐ **Returned** to owner / custodian

☐ **Transferred** to external agency or court

☐ **Destroyed** / securely wiped

DISPOSITION AUTHORIZED BY	DATE
Abdul Wahid	6 October 2026
REASON	**NEXT SCHEDULED ACTION**
Investigation ongoing — open items carried into later modules	Further module analysis & updated Chain of Custody